

AI Workflow Automation

AI Viral Reel Generator

Zero-Cost AI Systems · Vibe Coding · Prompt Engineering



Agenda

01

AI Tools & Industry Transformation

How modern AI tools are reshaping the developer landscape

02

AI Automation Use Cases

Real-world workflows automated across industries

03

Prompt Engineering

Crafting precise instructions to get reliable AI output

04

Vibe Coding

Building software through natural language intent

05

Hands-on: Reel Generator

Architecture and code walkthrough

Popular AI Tools for Developers

The modern AI toolkit — free, powerful, and accessible

▣ Claude

Reasoning & Code

Anthropic's assistant — excels at long-context reasoning, code, and safety.

💬 ChatGPT

Conversational AI

OpenAI's flagship model. Broad knowledge, plugins, and multimodal support.

✨ Gemini

Multimodal AI

Google DeepMind — strong at vision, data analysis, and Google integrations.

⚡ GitHub Copilot

Code Completion

AI pair programmer embedded in VS Code — autocomplete at line & function level.

📁 Cursor

AI Code Editor

Full IDE powered by AI — edit, refactor, and generate code via natural language.

🎬 Runway / Sora

Video Generation

Text-to-video AI. Generate clips, extend footage, and apply styles automatically.

AI Automation Use Cases

AI replacing repetitive work across every sector

Media & Content

Before Manual editing, scripting, subtitle work: 4–6 hrs per video

→ **After** AI pipeline: script → voiceover → B-roll → subtitles in under 60 sec

Healthcare

Before Doctors manually review patient records, write notes

→ **After** AI reads EHR data, flags anomalies, auto-drafts clinical notes

Software Engineering

Before Boilerplate code written by hand; manual bug hunts

→ **After** Copilot / Cursor generate functions, write tests, and explain errors

E-Commerce & Retail

Before Human agents handle every customer support ticket

→ **After** AI resolves 70% of tickets instantly — returns, tracking, FAQs

Prompt Engineering

The art and science of communicating with AI effectively



Be Specific

Vague prompts give vague results. Define role, context, format, and constraints explicitly.



Use Structure

Use delimiters, JSON schemas, or XML tags to force predictable structured output:
`response_mime_type="application/json"`



Iterate & Refine

Prompt engineering is iterative. Evaluate output, diagnose failure, then adjust tone, scope, or examples.

Anatomy of a Prompt

ROLE

You are a senior video content strategist.

CONTEXT

The user provides a raw topic sentence.

TASK

Split it into 3-7 second scene segments.

FORMAT

Return ONLY valid JSON:
`{ "scenes": [...] }`

CONSTRAINTS

No markdown, no explanation, max 7 scenes.

Vibe Coding

Writing software by describing intent, not syntax

"Express what you want the code to DO — the AI writes the HOW. You become the architect, the AI becomes the engineer."

Traditional Coding

✗ Write every line manually

✗ Debug syntax errors character by char

✗ Google Stack Overflow for every pattern

✗ Hours to scaffold a new service



Vibe Coding

✓ Describe intent: 'query Pexels for portrait video'



✓ AI fixes errors when you paste + describe the bug



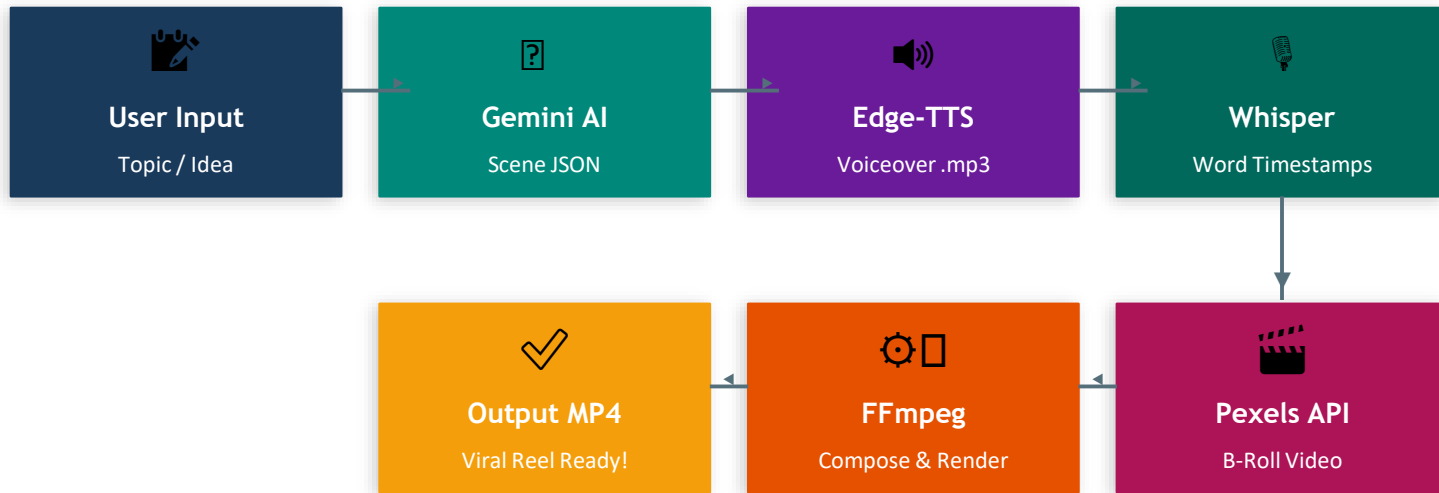
✓ Ask AI: 'how do I sidechain compress in FFmpeg?'



✓ Minutes: 'create a FastAPI endpoint that calls Gemini'

AI Viral Reel Generator – Architecture

How all components connect in a single automated pipeline



Gist

Text → JSON

JSON → Audio

Audio → Subs

Query → Video

Compose All

→ MP4 Output

Thank You

Questions, Code & Collaboration — Let's Build!



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